



12207 - NIRSpec mapping of the warm Disk Wind and icy envelope around an archetypal protostar

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
DG Tau B NIRSPEC G395H				
	2	DGTau-NIRSpec1	NIRSpec IFU Spectroscopy	(1) DG-TAU-B-NIRSPEC-1
	3	DGTau-NIRSpec2	NIRSpec IFU Spectroscopy	(2) DG-TAU-B-NIRSPEC-2

ABSTRACT

Disk winds launched by large scale magnetic fields have been recently suggested as major contributors to the transport of angular momentum and mass in proto-planetary disks. These magnetic winds may explain accretion in the low ionization regions of disks (the dead zone), still a major unsolved issue. JWST is playing a major role in the quest for observational evidence of these disk winds and the test of alternative explanations. The DG Tau B Class I protostar is a solar analog in the making, driving a rotating and velocity stratified cold CO outflow mapped with ALMA. Recent JWST NIRSpec observations identified a warm H₂ layer nested between the hot inner jet and the cold CO outflow. This nested outflow structure is

strongly reminiscent of the radial stratification expected in disk winds. However, the envelope contribution cannot be fully excluded. The central protostar also displays strong ice absorption signatures in its infrared spectrum. We propose to complement archival JWST NIRSpec and MIRI observations of this prototypical Class I source with currently missing NIRSpec-IFU data at 3-5 μm , allowing to map both the CO gas and major solids (CO_2 , H_2O) in a single setup. This full set of observations will provide stringent constraints on the radial stratification of the flow, critical to clarify its origin (MHD disk wind versus shocked cavity). Additionally, they will help establish the link between the natal envelope, disk and outflow solid components.

OBSERVING DESCRIPTION

We propose to map the central 6" x 6" of the DG Tau B outflow with NIRSpec-IFU in the 3-5 μm spectral range.

Instrument and observational strategy:

We will use NIRSpec-IFU with the grism G395H/F290 LP covering the wavelength range between 2.9 and 5.3 μm , which comprises all spectral signatures of interest in a single set-up (major ices + CO gas). Two IFU pointings will be obtained to cover both outflow lobes and the continuum position, similarly to archival program PID 1644. For each pointing and dithering position, we will obtain an MSA leakage exposure with the same detector parameters (but half total on source integration times). Indeed archival JWST NIRSpec observations (PID 1644) show that the bright nearby DG Tau star can induce leakage on the IFU exposures which may depend on the dithering position.

Exposure times

DG Tau B:

- Field centered on the redshifted lobe: We estimate ro-vibrational CO $v=1-0$ line brightness in the DG Tau B outflow using archival NIRSpec data on the $\text{H}_2(1-0)$ on both sources (PID 1644 & PID 1621) and assuming a similar CO/ $\text{H}_2(1-0)$ line ratio to be between 2.5×10^{17} to $2.5 \times 10^{16} \text{ erg/s/cm}^2/\text{arcsec}^2$ for distances to the source ≤ 1 . We use the ETC with a source of uniform brightness if a CO line (of width 20 km/s) at 4.7 μm of $2.5 \times 10^{17} \text{ erg/s/cm}^2/\text{arcsec}^2$. With the NRSIRS2RAPID readout mode, 2 integrations of 30 groups

and 4 point dithering (for extended source), we reach a SNR per spaxel ~ 5 for the fainter CO(1-0) transitions (~ 50 for the brighter ones).

- for the field centered on the protostar and blue lobe: from the archival JWST NIRSpec and MIRI-MRS data (PID 1644), we estimate the brightness of the central source between 0.05 Jy at 3.5 μ m and 0.2 Jy at 5 μ m. We will employ the NRSIRS2RAPID readout pattern, with 4 integrations of 7 groups. We will use a 9pt cycling dithering pattern with medium offset to ensure better sampling of the PSF. This will ensure a SNR ~ 500 on the continuum on source (and SNR > 10 per spaxel in the extended scattered light contribution on scales of a few hundreds of au) while only partially saturating at the end of the ramp towards the end of the spectral range for a few central spaxels.

Proposal 12207 - Targets - NIRSPEC mapping of the warm Disk Wind and icy envelope around an archetypal protostar

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	DG-TAU-B-NIRSPEC-1	RA: 04 27 2.6067 (66.7608613d) Dec: +26 05 29.93 (26.09165d) Equinox: J2000		
	Comments: Category=Star Description=[Protostars] Extended=YES				
Fixed Targets	(2)	DG-TAU-B-NIRSPEC-2	RA: 04 27 2.4223 (66.7600929d) Dec: +26 05 31.22 (26.09201d) Equinox: J2000		
	Comments: Category=Star Description=[Protostars] Extended=YES				

Proposal 12207 - Observation 2 - NIRSpec mapping of the warm Disk Wind and icy envelope around an archetypal protostar

Observation	Proposal 12207, Observation 2: DGTau-NIRSpec1 Fri Mar 13 22:07:51 GMT 2026											
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	DG-TAU-B-NIRSPEC-1	RA: 04 27 2.6067 (66.7608613d) Dec: +26 05 29.93 (26.09165d) Equinox: J2000									
Template	<i>Comments:</i> <i>Category=Star</i> <i>Description=[Protostars]</i> <i>Extended=YES</i>											
	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		MEDIUM		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	7	4	false	true	WAVECAL	9	36	4201.6	
	2	G395H/F290LP	NRSIRS2RAPID	7	2	true	true	NONE	9	18	2100.8	
Special Requirements	No Parallel Attachments											

Proposal 12207 - Observation 3 - NIRSpec mapping of the warm Disk Wind and icy envelope around an archetypal protostar

Observation	Proposal 12207, Observation 3: DGTau-NIRSpec2 Fri Mar 13 22:07:51 GMT 2026											
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	DG-TAU-B-NIRSPEC-2	RA: 04 27 2.4223 (66.7600929d) Dec: +26 05 31.22 (26.09201d) Equinox: J2000									
Template	<i>Comments:</i> <i>Category=Star</i> <i>Description=[Protostars]</i> <i>Extended=YES</i>											
	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	30	2	false	true	WAVECAL	4	8	3618.045	
	2	G395H/F290LP	NRSIRS2RAPID	30	1	true	true	NONE	4	4	1809.022	
Special Requirements	No Parallel Attachments											